

SANTOSH PANT

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EDUCATION

Knox College

September 2023 - June 2027

Bachelor of Science in Computer Science and Data Science (GPA: 3.82)

Galesburg, Illinois

- **Honors:** Break Through Tech AI Fellow (Cornell Tech, 2025-26); Richter Research Scholar; Dean's List (6/6 terms); ICPC North Central Participant (UIUC site, 2025); Teaching Assistant - Data Structures & Algorithms; CS Club Lead
- **Hackathons:** UMich AI for Heat Resilience - 2nd Place (\$2K); HackIllinois 2026 (UIUC) - Best Use of Action VectorAI DB (1st, sponsor track); ASA DataFest 2026 - Best Insights Track; HackAugie 2026 - Best Gaming Hack

GEOSPATIAL ML RESEARCH EXPERIENCE

Machine Learning Research Collaborator

June 2026 - Present

RGB-to-Thermal Urban Heat Mapping - Mentor: Prof. Geoffrey Siwo, UMich Center for Global Health Equity

Remote

- Predicting street-level urban heat maps from RGB drone imagery (no thermal camera) with the hackathon's organizing professor, toward cheap, scalable heat-risk mapping (github.com/thesantoshpant/rgb-to-thermal).
- Diagnosed two silent data faults the hackathon pipeline missed (colormapped-JPEG target needing palette inversion; RGB/thermal misalignment from sensor field-of-view mismatch) and rebuilt the pipeline (ConvNeXt U-Net, pix2pix cGAN); the 2nd-place hackathon model scored 19.3 dB PSNR / 0.66 SSIM on a 202-image held-out set. Targeting WACV 2027.

Satellite Heat-Stress Forecasting, Nepal Terai

2026 - Present

Mentor: Prof. Andrew Leahy, Knox College (first-author paper under review at ACM SIGSPATIAL 2026)

Galesburg, Illinois

- Built an early-season vegetation heat-stress pipeline over 21 Terai rice districts using Google Earth Engine (MODIS LST/NDVI, ERA5-Land, ESA WorldCover; 59,793 pixel-years); deployed a live forecast app (terai-heat-forecaster.streamlit.app).
- Showed previous-year persistence beats a tuned 15-feature XGBoost on average (macro-F1 0.476 vs 0.362) under year-blocked leave-one-district-out evaluation; proposed persistence + modal-class nulls as required baselines.

Geospatial Foundation-Model Benchmark (Frozen vs. Fine-Tuning)

2026 - Present

Independent researcher, Knox College (under review, TMLR 2026)

Galesburg, Illinois

- Controlled evaluation of geospatial foundation models (Prithvi-EO-2.0, TerraMind) for field-extent segmentation across six countries (Fields of The World); a frozen decoder is statistically equivalent to full fine-tuning in 9 of 12 cells at ~110x fewer parameters, and proxy land-cover labels inflate weak baselines by up to +0.34 AUROC.

Conformal Prediction Intervals for District Paddy-Yield Loss, Nepal

2026 - Present

Independent researcher, Knox College (under review, SIGSPATIAL 2026 Applications)

Galesburg, Illinois

- Built leak-controlled district-level 90% conformal prediction intervals for Nepal's 2025 Madhesh drought (covariate-shift-weighted conformal + a finite-sample coverage theorem), with an operational decision-support classifier that abstains when calibration is insufficient.

Cross-Region Fire-Attribution Audit (Nepal Terai, Punjab, MP)

2026 - Present

Independent researcher, Knox College

Galesburg, Illinois

- Audited MODIS/VIIRS fire detections used as crop-residue proxies; across four cropland masks, the share of rabi detections outside cropland rises from Punjab (4.5%) to Madhya Pradesh (18%) to Nepal Terai (98%), validated against published DARBE numbers within 1.24x.

ENGINEERING EXPERIENCE

Software Engineering Intern - The MACRO Consortium

March 2025 - June 2025

- Shipped a priority-based scheduling algorithm into the open-source pyscope astronomy package; investigated and mitigated a data-loss bug, improving reliability for nightly observation batches at Knox College and partner institutions.

SELECTED PROJECTS

CropScan | PyTorch (DINOv2 ViT + LoRA, EfficientNetV2), FastAPI, React, MongoDB, Docker

cropsan.tech

- Full-stack crop-disease app for smallholder farmers: a calibrated two-model ensemble (DINOv2 ViT + LoRA and EfficientNetV2) that abstains when the models disagree, plus AI treatment guidance; shipped with auth, Docker, and CI.

SKILLS

- **Geospatial & Remote Sensing:** Google Earth Engine, GIS, Rasterio, geopandas, Sentinel-1/2, MODIS, Landsat, PlanetScope, ERA5, ESA WorldCover, AlphaEarth, drone imagery
- **ML & AI:** PyTorch, deep learning, computer vision, scikit-learn, XGBoost, EO foundation models (Clay, Prithvi, TerraMind, AnySat), CNNs, U-Net, Grad-CAM
- **Languages:** Python, R, TypeScript, Go, SQL
- **Tools:** FastAPI, Streamlit, Folium, PostgreSQL, AWS, Docker, Git, Linux, cloud data pipelines